

Surgical VR - User Guide

Supported Device

This application is designed exclusively for Meta Quest 3 and will not function on other devices.

Getting Started

1. Setting Up the Boundary

Before using the application, you need to set up a Room-Scale or Stationary Boundary to define your play area. It is recommended to use Room-Scale with a larger boundary for better functionality and movement.

How to Set the Boundary:

- Click on the bottom-left menu bar in your Meta Quest 3 interface.
- Open Menu Options and select Boundary.
- Choose either Room-Scale or Stationary based on your preference.
- Recommended: Select Room-Scale and draw a large boundary for better movement and testing other features.
- Follow the official Meta Quest guide on boundary setup: [Click Here](#)

 **Note:** If you move beyond the set boundary, the application will not display the output.

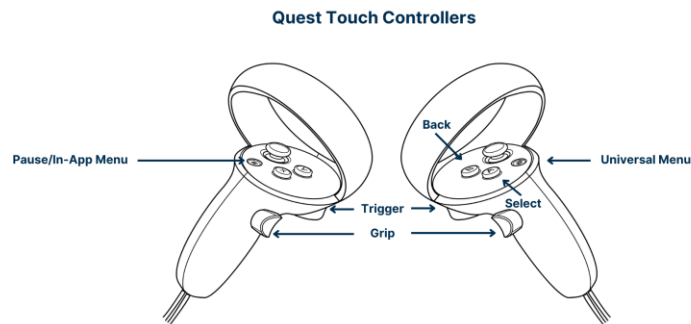
Application Features & Controls

2. Interacting with the Quad (Virtual Object)

2.1 Moving the Quad (Using Controllers)

Press the 🎮 Trigger Button + Grip Button together to create a fist.

- Pull towards you or away to move the quad forward and backward.
- Slide left or right to move horizontally.
- This action can be performed with either the left or right controller.



2.2 Scaling the Quad (Using Controllers)

- Press and hold 🎮 Trigger + Grip Button on both controllers.
- Move diagonally to scale up or scale down the quad.

2.3 Rotating the Quad (Using Controllers)

- Press and hold 🎮 Trigger + Grip Button on any one controller.
- spin towards up direction or down direction to rotate the quad.

2.4 Interacting with the Quad (Using Hands - Hand Tracking)

- To use hand tracking, place your controllers on a table or on the ground.
- Show your hands in front of the headset to activate virtual hands.

To Move the Quad:

- Pinch (Thumb + Index Finger) Gesture with either hand to grab and move the quad.

To Scale the Quad:

- Pinch Gesture with both hands and move outward to zoom in or inward to zoom out.

To Rotate the Quad:

- Pinch Gesture with one hand and spin in up or down direction to rotate the quad.

3. Application UI & View Controls

3.1 Minimizing and Maximizing the Window

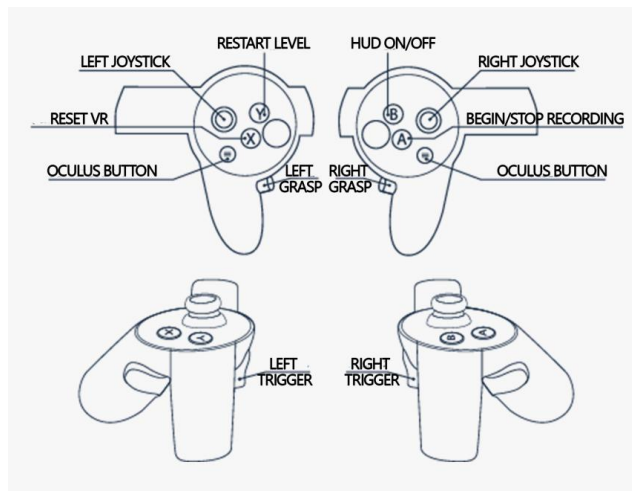
- Minimize the window: Press the Left Thumbstick Button (Joystick Click) on the left controller.
- Maximize the window: Press the same button again.

3.2 Switching Between Virtual and Passthrough Mode

- Switch between VR and Passthrough Mode: Press the Y Button on the left controller.
- In Virtual Mode, you can still manipulate the quad as described above.

3.3 Re-Centering the View

- Using the Right Controller: Press the Oculus Button on the right controller to reset the view.
- Using Hand Tracking: Make a Pinch Gesture (Thumb + Index Finger) on your right hand to re-center the view.



💡 **Note:** Minimization and switching between VR/Passthrough mode are only available via controllers, not hand gestures.

Final Notes

- Ensure your play area is free from obstacles to move safely.
- If you experience tracking issues, check the room lighting and recalibrate the boundary.